

Special Matrices and Transposition

- **Definition 1**

The **identity matrix** $I_n = [\delta_{ij}]_n \in M_n(S)$ is defined by

$$\delta_{ij} = \begin{cases} 1 & (i = j) \\ 0 & (i \neq j) \end{cases}.$$

- **Example 1.1**

$$I_2 = \begin{bmatrix} \delta_{11} & \delta_{12} \\ \delta_{21} & \delta_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- **Proposition 1.2**

For every $A \in M_{m \times n}(S)$

$$I_m A = A \quad \text{and} \quad A I_n = A.$$

- **Proof 1.2.1**

Let $A = [a_{ij}]_{m \times n}$ then

$$\begin{aligned} I_m A &= [c_{ij}]_{m \times n} \\ &= [a_{ij}]_{m \times n}. \end{aligned}$$

where $c_{ij} = \sum_{k=1}^m \delta_{ik} a_{kj} = \delta_{ii} a_{ij} = a_{ij}$.

$$\begin{aligned} A I_n &= [c_{ij}]_{m \times n} \\ &= [a_{ij}]_{m \times n}. \end{aligned}$$

where $c_{ij} = \sum_{k=1}^m a_{ik} \delta_{kj} = a_{ij} \delta_{jj} = a_{ij}$.

- **Definition 2**

A matrix $A \in M_n(S)$ is called **invertible** if there exists $B \in M_n(S)$ such that

$$AB = I_n = BA.$$

If it exists, B is called the **inverse** of A and is denoted by A^{-1} .

- **Theorem 2.1**

If $A \in M_n(S)$ is an invertible matrix then its inverse is unique. Moreover, if A and B are invertible matrices with the same size, then AB is also an invertible matrix and $(AB)^{-1} = B^{-1}A^{-1}$.

- **Proof 2.1.1**

Suppose that B and C are inverses of A then

$$AB = I_n = BA, \quad AC = I_n = CA.$$

Hence

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

Therefore the inverse is unique. Now, since

$$AB(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_n A^{-1} = AA^{-1} = I_n.$$

and

$$(B^{-1}A^{-1})AB = B^{-1}(A^{-1}A)B = B^{-1}I_n B = B^{-1}B = I_n.$$

we have the AB is an invertible matrix and $(AB)^{-1} = B^{-1}A^{-1}$ because the inverse is unique.

◦ **Example 2.2**

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad - bc \neq 0$ then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

In fact, let $B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$ such that $AB = I_2$ then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} ax + bz & ay + bw \\ cx + dz & cy + dw \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

so we obtain the linear system

$$\begin{cases} ax + bz = 1 & (E_1) \\ ay + bw = 0 & (E_2) \\ cx + dz = 0 & (E_3) \\ cy + dw = 1 & (E_4) \end{cases}$$

From $d(E_1) - b(E_3)$, $d(E_2) - b(E_4)$, $c(E_1) - a(E_3)$, $c(E_2) - a(E_4)$ we obtain the following equation

$$\begin{cases} (ad - bc)x = d \\ (ad - bc)y = -b \\ (ad - bc)z = -c \\ (ad - bc)w = a \end{cases}$$

Now, if $ad - bc \neq 0$ we have that

$$\begin{cases} x = \frac{1}{ad - bc}d \\ y = \frac{1}{ad - bc}(-b) \\ z = \frac{1}{ad - bc}(-c) \\ w = \frac{1}{ad - bc}a \end{cases}$$

Hence

$$B = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

with $AB = I_2 = BA$. Therefore A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

◦ **Example 2.3**

Determine if the matrix $A = \begin{bmatrix} 2 & 3 \\ 2 & 2 \end{bmatrix}$ is invertible and, if so, find its inverse.

■ **Solution 2.3.1**

Since $2 \times 2 - 3 \times 2 = -2 \neq 0$ then A is invertible and

$$A^{-1} = -\frac{1}{2} \begin{bmatrix} 2 & -3 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} -1 & \frac{3}{2} \\ 1 & -1 \end{bmatrix}.$$

◦ **Remark 2.4**

Note that $A \in M_2(\mathbb{Z})$ but $A^{-1} \notin M_2(\mathbb{Z})$.

• **Definition 3**

A matrix $D = [d_{ij}]_n \in M_n(S)$ is **diagonal** if $d_{ij} = 0$ for $i \neq j$ and we write $D = \text{diag}(d_{11}, d_{22}, \dots, d_{nn})$. That is

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 & 0 \\ 0 & d_{22} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & d_{n-1n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_{nn} \end{bmatrix}.$$

◦ **Remark 3.1**

If $D, E \in M_n(S)$ are diagonal matrices then

$$\text{diag}(d_{11}, d_{22}, \dots, d_{nn}) \text{diag}(e_{11}, e_{22}, \dots, e_{nn}) = \text{diag}(d_{11}e_{11}, d_{22}e_{22}, \dots, d_{nn}e_{nn}).$$

Hence diagonal matrices commute.

◦ **Remark 3.2**

Multiplication by a diagonal matrix has a simple effect. If $A \in M_n(S)$ and $D = \text{diag}(d_{11}, d_{22}, \dots, d_{nn})$ then AD is obtained by multiplying column j of A by d_{jj} . Similarly, DA is obtained by multiplying the row i of A by d_{ii} . For example

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix} = \begin{bmatrix} a_{11}d_{11} & a_{12}d_{22} & a_{13}d_{33} \\ a_{21}d_{11} & a_{22}d_{22} & a_{23}d_{33} \\ a_{31}d_{11} & a_{32}d_{22} & a_{33}d_{33} \end{bmatrix}.$$

• **Definition 4**

A matrix $U = [u_{ij}] \in M_n(S)$ is **upper triangular** if $u_{ij} = 0$ for $i > j$.

A matrix $L = [l_{ij}] \in M_n(S)$ is **lower triangular** if $l_{ij} = 0$ for $i < j$.

◦ **Example 4.1**

Let

$$A = \begin{bmatrix} 2 & -3 & 4 \\ 0 & 5 & 1 \\ 0 & 0 & -7 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ -4 & -3 & 0 \\ 8 & 9 & 6 \end{bmatrix}.$$

Then A is upper triangular and B is lower triangular.

◦ **Proposition 4.2**

The product of two upper triangular matrices with the same size is upper triangular.

The product of two lower triangular matrices with the same size is lower triangular.

■ **Proof 4.2.1**

Let $A, B \in M_n(S)$ be upper triangular matrices then

$$AB = [c_{ik}]_n \text{ where } c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}.$$

Let $i > k$. If $i > j$ then $a_{ij} = 0$ because A is upper triangular and if $j \geq i > k$ then $b_{jk} = 0$ because B is upper triangular. Hence

$$c_{ik} = \sum_{j=1}^n a_{ij}b_{jk} = 0$$

and thus AB is upper triangular. Analogously, the product of two lower triangular matrices with the same size is lower triangular.

• **Definition 5**

Let $A = [a_{ij}]_{m \times n} \in M_{m \times n}(S)$. Then the **transpose** of A , denoted by $A^\top$, is the matrix obtained by interchanging the rows and the columns of A , that is, $A^\top = [a_{ji}]_{n \times m} \in M_{n \times m}(S)$.

◦ **Example 5.1**

If

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 5 & -6 \end{bmatrix}$$

then

$$A^\top = \begin{bmatrix} 1 & 4 \\ -2 & 5 \\ 3 & -6 \end{bmatrix}.$$

◦ **Theorem 5.2**

Let A, B be matrices with sizes such that the following operations make sense and let $\alpha \in S$.

1. $(A^\top)^\top = A$.
2. $(A + B)^\top = A^\top + B^\top$.
3. $(\alpha A)^\top = \alpha A^\top$.
4. $(AB)^\top = B^\top A^\top$.

■ **Proof 5.2.1**

1. Let $A = [a_{ij}]_{m \times n}$ then

$$(A^\top)^\top = ([a_{ji}]_{n \times m})^\top = [a_{ij}]_{m \times n} = A.$$

2. Let $A = [a_{ij}]_{m \times n}, B = [b_{ij}]_{m \times n}$ then

$$\begin{aligned} (A + B)^\top &= ([a_{ij} + b_{ij}]_{m \times n})^\top \\ &= [a_{ji} + b_{ji}]_{n \times m} \\ &= [a_{ji}]_{n \times m} + [b_{ji}]_{n \times m} \\ &= A^\top + B^\top. \end{aligned}$$

3. Let $A = [a_{ij}]_{m \times n}$ then

$$\begin{aligned} (\alpha A)^\top &= ([\alpha a_{ij}]_{m \times n})^\top \\ &= [\alpha a_{ji}]_{n \times m} \\ &= \alpha [a_{ji}]_{n \times m} \\ &= \alpha A^\top. \end{aligned}$$

4. Let $A = [a_{ij}]_{m \times n}, B = [b_{jk}]_{n \times p}$ then

$$\begin{aligned}
(AB)^\top &= \left([c_{ik}]_{m \times p}\right)^\top & \text{where } c_{ik} &= \sum_{j=1}^n a_{ij}b_{jk} \\
&= [c_{ki}]_{p \times m} & \text{where } c_{ki} &= \sum_{j=1}^n b_{kj}a_{ji} \\
&= [b_{kj}]_{p \times n} [a_{ji}]_{n \times m} \\
&= B^\top A^\top.
\end{aligned}$$

◦ **Definition 6**

Let $A \in M_n(S)$.

A is called **symmetric** if $A^\top = A$, that is, if $a_{ji} = a_{ij}$ for all $1 \leq i, j \leq n$.

A is called **antisymmetric** if $A^\top = -A$, that is, if $a_{ji} = -a_{ij}$ for all $1 \leq i, j \leq n$.

■ **Remark 6.1**

If A is antisymmetric then the entries on the diagonal $a_{11}, a_{22}, \dots, a_{nn}$ are 0. Since $a_{ii} = -a_{ii}$, that is, $2a_{ii} = 0$ and thus $a_{ii} = 0$ for all $1 \leq i \leq n$.

■ **Example 6.2**

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 0 & 4 \\ 3 & -4 & 0 \end{bmatrix}.$$

Then A is symmetric and B is antisymmetric.

■ **Proposition 6.3**

1. If $A \in M_n(S)$ then $A + A^\top$ is symmetric and $A - A^\top$ is antisymmetric.
2. If $A \in M_{m \times n}(S)$ then both $AA^\top$ and $A^\top A$ are symmetric.
3. If A and B are symmetric matrices of the same size then AB is symmetric, if and only if, $AB = BA$.

■ **Proof 6.3.1**

$$\begin{aligned}
1. \quad (A + A^\top)^\top &= A^\top + (A^\top)^\top \\
&= A^\top + A \\
&= A + A^\top.
\end{aligned}$$

Therefore $A + A^\top$ is symmetric.

$$\begin{aligned}
(A - A^\top)^\top &= A^\top + (-A^\top)^\top \\
&= A^\top - (A^\top)^\top \\
&= A^\top - A \\
&= -(A - A^\top).
\end{aligned}$$

Therefore $A - A^\top$ is antisymmetric.

$$\begin{aligned}
2. \quad (AA^\top)^\top &= (A^\top)^\top A^\top \\
&= AA^\top
\end{aligned}$$

Therefore $AA^\top$ is symmetric.

$$\begin{aligned}
(A^\top A)^\top &= A^\top (A^\top)^\top \\
&= A^\top A
\end{aligned}$$

Therefore $A^\top A$ is symmetric.

3. ($\implies$) Suppose that AB is symmetric then

$$(AB)^\top = AB$$

$$B^\top A^\top = AB$$

$$BA = AB$$

Because A, B are symmetric.

($\impliedby$) Suppose that $AB = BA$ hence

$$(AB)^\top = B^\top A^\top$$

$$= BA$$

$$= AB$$

Because A, B are symmetric. Therefore AB is symmetric.